KAIZEN MATHS

Lesson starter

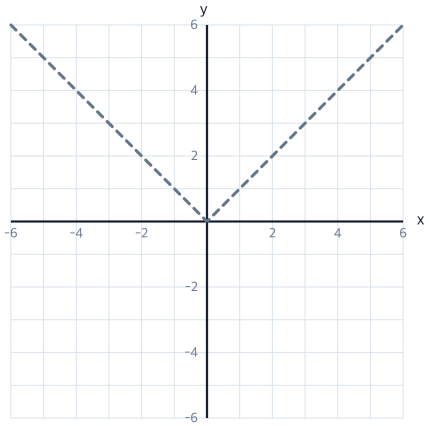
Graph Transformations and Curve Sketching

Single Transformations

Name _____

SA1

Sketch the image graph after the vertical translation.



y = f(x)to be sketched

The dashed graph is f(x) = |x|. Sketch y = f(x) + 2 and describe the transformation.

Show your working

KAIZEN MATHS

Lesson starter

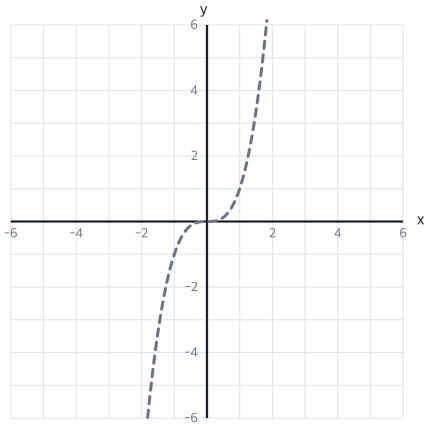
Graph Transformations and Curve Sketching

Single Transformations

Name _____

SB1

Sketch the image graph after the horizontal translation.



y = f(x)to be sketched

The dashed graph is $f(x) = x^3$. Sketch y = f(x - 2) and describe the transformation.

Show your working

KAIZEN MATHS

Lesson starter

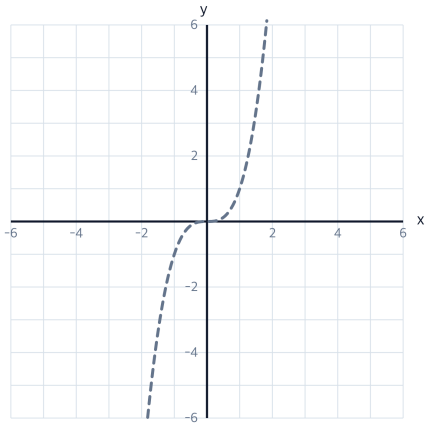
Graph Transformations and Curve Sketching

Single Transformations

Name _____

SC1

Sketch the image graph after the reflection.



y = f(x)to be sketched

The dashed graph is $f(x) = x^3$. Sketch $y = f(-x)$ and describe the transformation.

ALG-146D-L2 · SC

Show your working

KAIZEN MATHS

Lesson starter

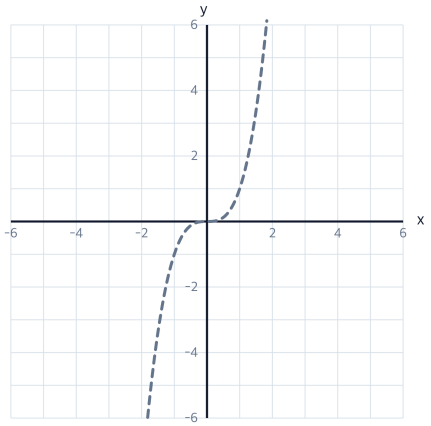
Graph Transformations and Curve Sketching

Single Transformations

Name _____

SD1

Sketch the image graph after the stretch.



y = f(x)to be sketched

The dashed graph is $f(x) = x^3$. Sketch $y = 2f(x)$ and describe the transformation.

ALG-146D-L2 · SD

Show your working

KAIZEN MATHS

Exit ticket

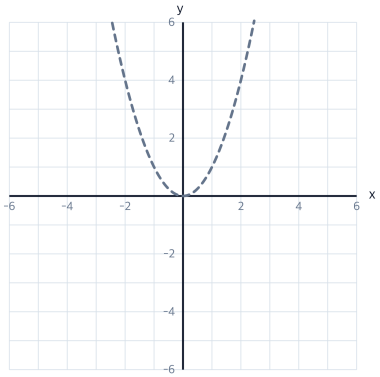
Graph Transformations and Curve Sketching

Single Transformations

Name

EA1

Sketch the image graph after the vertical translation.



y = f(x)to be sketched

The dashed graph is $f(x) = x^2$. Sketch $y = f(x) - 3$ and describe the transformation.

ALG-146D-L2 · EA

Show your working

KAIZEN MATHS

Exit ticket

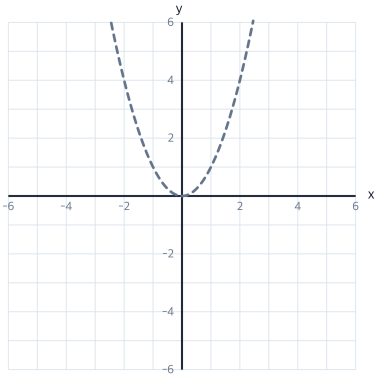
Graph Transformations and Curve Sketching

Single Transformations

Name

EB1

Sketch the image graph after the horizontal translation.



y = f(x)to be sketched

The dashed graph is $f(x) = x^2$. Sketch $y = f(x - 3)$ and describe the transformation.

ALG-146D-L2 · EB

Show your working

KAIZEN MATHS

Exit ticket

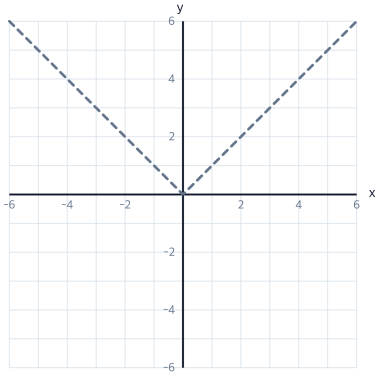
Graph Transformations and Curve Sketching

Single Transformations

Name

EC1

Sketch the image graph after the reflection.



y = f(x)to be sketched

The dashed graph is $f(x) = |x|$. Sketch $y = -f(x)$ and describe the transformation.

ALG-146D-L2 · EC

Show your working

KAIZEN MATHS

Exit ticket

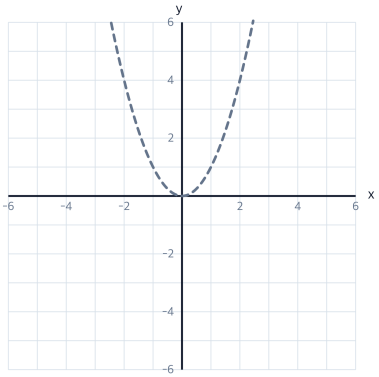
Graph Transformations and Curve Sketching

Single Transformations

Name

EA1

Sketch the image graph after the vertical translation.



y = f(x)to be sketched

The dashed graph is $f(x) = x^2$. Sketch $y = f(x) - 3$ and describe the transformation.

ALG-146D-L2 · EA

Show your working

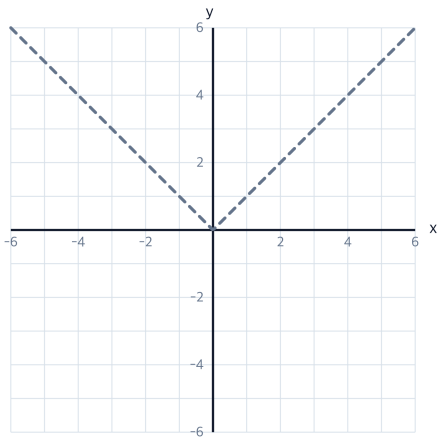
Teacher answers & working

Graph Transformations and Curve Sketching · Single Transformations

ALG-146D-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1

Sketch the image graph after the vertical translation.



y = f(x)to be sketched

The dashed graph is f(x) = |x|. Sketch y = f(x) + 2 and describe the transformation.

Answer:

Translation 2 units up.

Working:

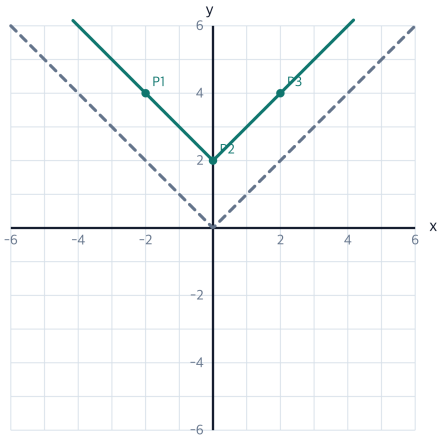
Start with the parent graph f(x) = |x|.

The equation y = f(x) + 2 gives the point mapping (x, y) → (x, y + 2).

Map useful points on the original graph.

Original point	Image point
(-2, 2)	(-2, 4)
(0, 0)	(0, 2)
(2, 2)	(2, 4)

Draw a smooth image curve through the image points, keeping the shape of the parent graph consistent.



y = f(x)y = f(x) + 2

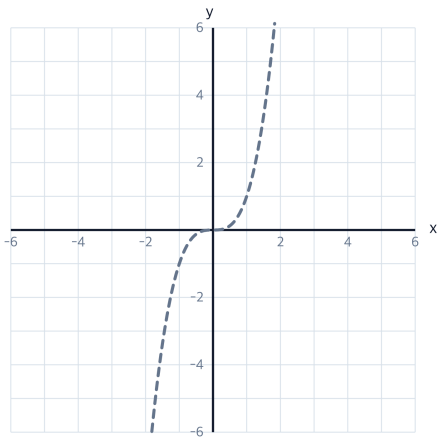
Teacher answers & working

Graph Transformations and Curve Sketching · Single Transformations

ALG-146D-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1

Sketch the image graph after the horizontal translation.



y = f(x)to be sketched

The dashed graph is $f(x) = x^3$. Sketch $y = f(x - 2)$ and describe the transformation.

Answer:

Translation 2 units right.

Working:

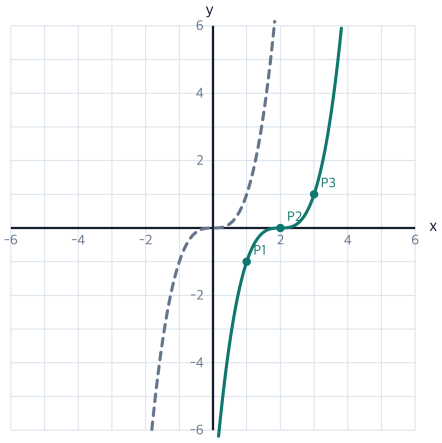
Start with the parent graph $f(x) = x^3$.

The equation $y = f(x - 2)$ gives the point mapping $(x, y) \rightarrow (x + 2, y)$.

Map useful points on the original graph.

Original point	Image point
(-1, -1)	(1, -1)
(0, 0)	(2, 0)
(1, 1)	(3, 1)

Draw a smooth image curve through the image points, keeping the shape of the parent graph consistent.



y = f(x)y = f(x - 2)

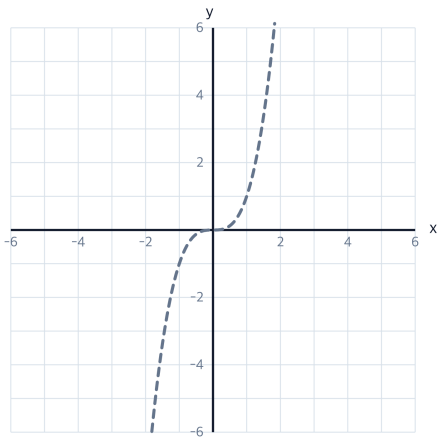
Teacher answers & working

Graph Transformations and Curve Sketching · Single Transformations

ALG-146D-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1

Sketch the image graph after the reflection.



y = f(x)to be sketched

The dashed graph is $f(x) = x^3$. Sketch y = f(-x) and describe the transformation.

Answer:

Reflection in the y-axis.

Working:

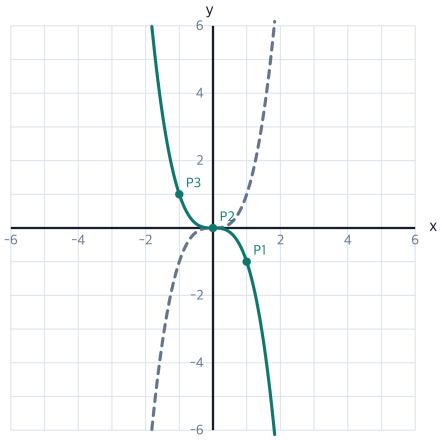
Start with the parent graph $f(x) = x^3$.

The equation y = f(-x) gives the point mapping (x, y) → (-x, y).

Map useful points on the original graph.

Original point	Image point
(-1, -1)	(1, -1)
(0, 0)	(0, 0)
(1, 1)	(-1, 1)

Draw a smooth image curve through the image points, keeping the shape of the parent graph consistent.



y = f(x)y = f(-x)

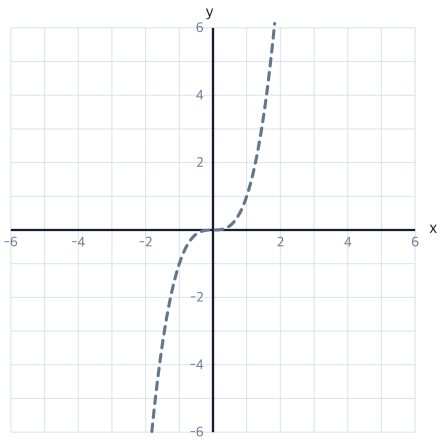
Teacher answers & working

Graph Transformations and Curve Sketching · Single Transformations

ALG-146D-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1

Sketch the image graph after the stretch.



y = f(x)to be sketched

The dashed graph is $f(x) = x^3$. Sketch $y = 2f(x)$ and describe the transformation.

Answer:

Vertical stretch by scale factor 2.

Working:

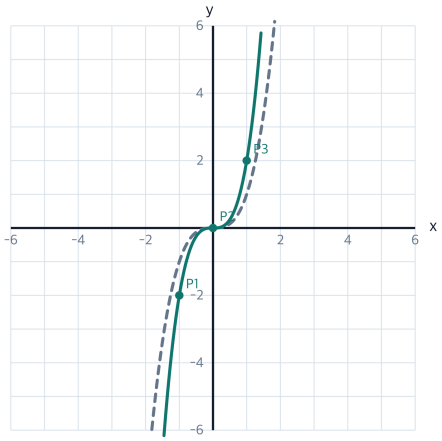
Start with the parent graph $f(x) = x^3$.

The equation $y = 2f(x)$ gives the point mapping $(x, y) \rightarrow (x, 2y)$.

Map useful points on the original graph.

Original point	Image point
(-1, -1)	(-1, -2)
(0, 0)	(0, 0)
(1, 1)	(1, 2)

Draw a smooth image curve through the image points, keeping the shape of the parent graph consistent.



$y = f(x) y = 2f(x)$

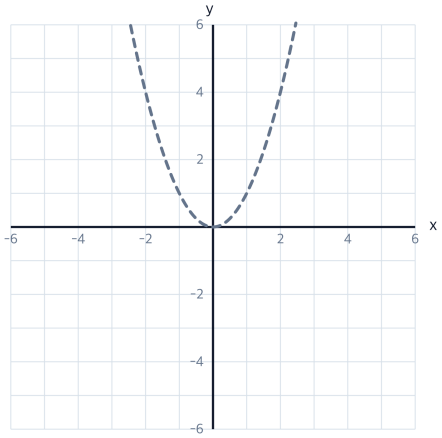
Teacher answers & working

Graph Transformations and Curve Sketching · Single Transformations

ALG-146D-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1

Sketch the image graph after the vertical translation.



y = f(x)to be sketched

The dashed graph is $f(x) = x^2$. Sketch $y = f(x) - 3$ and describe the transformation.

Answer:

Translation 3 units down.

Working:

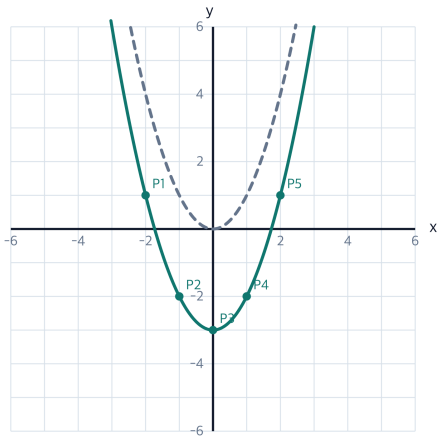
Start with the parent graph $f(x) = x^2$.

The equation $y = f(x) - 3$ gives the point mapping $(x, y) \rightarrow (x, y - 3)$.

Map useful points on the original graph.

Original point	Image point
(-2, 4)	(-2, 1)
(-1, 1)	(-1, -2)
(0, 0)	(0, -3)
(1, 1)	(1, -2)
(2, 4)	(2, 1)

Draw a smooth image curve through the image points, keeping the shape of the parent graph consistent.



Teacher answers & working

Graph Transformations and Curve Sketching · Single Transformations

ALG-146D-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1 (continued)

$y = f(x) y = f(x) - 3$

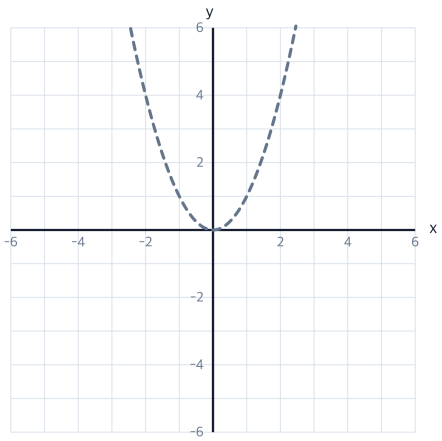
Teacher answers & working

Graph Transformations and Curve Sketching · Single Transformations

ALG-146D-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1

Sketch the image graph after the horizontal translation.



y = f(x)to be sketched

The dashed graph is $f(x) = x^2$. Sketch y = f(x - 3) and describe the transformation.

Answer:

Translation 3 units right.

Working:

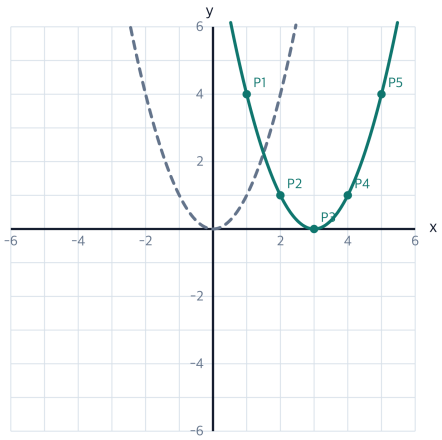
Start with the parent graph $f(x) = x^2$.

The equation y = f(x - 3) gives the point mapping (x, y) → (x + 3, y).

Map useful points on the original graph.

Original point	Image point
(-2, 4)	(1, 4)
(-1, 1)	(2, 1)
(0, 0)	(3, 0)
(1, 1)	(4, 1)
(2, 4)	(5, 4)

Draw a smooth image curve through the image points, keeping the shape of the parent graph consistent.



Teacher answers & working

Graph Transformations and Curve Sketching · Single Transformations

ALG-146D-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1 (continued)

$y = f(x) y = f(x - 3)$

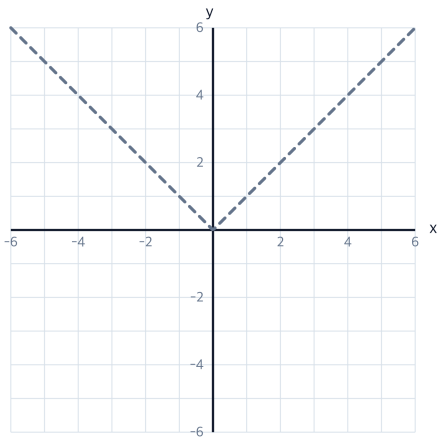
Teacher answers & working

Graph Transformations and Curve Sketching · Single Transformations

ALG-146D-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1

Sketch the image graph after the reflection.



y = f(x)to be sketched

The dashed graph is f(x) = |x|. Sketch y = -f(x) and describe the transformation.

Answer:

Reflection in the x-axis.

Working:

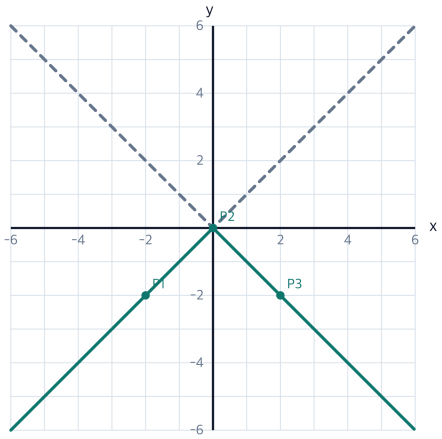
Start with the parent graph f(x) = |x|.

The equation y = -f(x) gives the point mapping (x, y) → (x, -y).

Map useful points on the original graph.

Original point	Image point
(-2, 2)	(-2, -2)
(0, 0)	(0, 0)
(2, 2)	(2, -2)

Draw a smooth image curve through the image points, keeping the shape of the parent graph consistent.



y = f(x)y = -f(x)